Indian Statistical Institute

Kolkata Centre

Bachelor of Statistics and Data Science (BSDS)

Optimization Using Numerical Methods

Electricity Bill Minimization via Linear Programming

Submitted by	Ankesh Raj Aniket (BSDCC2502)
	Md Danish (BSDCC2510)
	Dunga Pardha Saradhi (BSDCC2506)
Course Instructor	Kaushik Jana

Dataset	Kaggle Dataset
GitHub Repository	GitHub Repository

April 2026

Abstract

This report formulates the household electricity bill minimisation problem as a linear programme (LP) under a two-period peak/off-peak tariff structure. Appliance-level annual demands are computed from a Kaggle dataset of 45,345 household-month observations across six appliances and twelve calendar months. Using an average observed tariff $\bar{c} = 8.3696$ Rs./kWh, peak and off-peak rates of $1.2\bar{c}$ and $0.8\bar{c}$ are constructed. The LP is solved in `R` using the `lpSolve` package. We prove analytically via KKT conditions that the unique optimal solution satisfies $x_i^* = R_i$ for all appliances, and that the optimised cost equals exactly $\frac{2}{3}$ of the baseline, guaranteeing a saving of $\frac{1}{3} \approx 33.3\%$. A stochastic extension with random tariff $C \sim \mathcal{N}(\mu, \sigma^2)$ confirms the result holds in expectation. The annual bill falls from **Rs. 2,39,40,746** to **Rs. 1,59,60,497**, saving **Rs. 79,80,249** at an efficiency of **Rs. 3.35 per kWh**.

1 Problem

1.1 Research Problem (RP)

Household electricity bills in India depend not only on total energy consumed but also on *when* that energy is consumed. Grid operators distinguish between *peak hours* (roughly 18:00–22:00, when aggregate demand is high and unit cost is elevated) and *off-peak hours* (23:00–06:00, when the grid is lightly loaded and tariffs are lower). If an appliance whose operation is time-flexible—a fan, television, or motor pump—can be rescheduled to off-peak windows, the household pays for the same energy at a lower unit price, reducing the bill without any sacrifice of comfort or service quality. This creates a structured resource allocation problem amenable to linear programming.

1.2 Research Question (RQ)

By how much can the annual electricity bill of a household be reduced if appliance usage is optimally reallocated from peak-hour to off-peak-hour tariff windows using linear programming, and what is the per-unit savings efficiency of this reallocation?

2 Plan

2.1 Data Source and Sampling

The dataset is a secondary, observational cross-section obtained from Kaggle. Each row records one household-month observation with appliance-level usage (kWh), city name, electricity company, monthly hours of operation, prevailing tariff rate (Rs./kWh), and the resulting electricity bill (Rs.). Records span multiple Indian cities—Mumbai, Hyderabad,

Vadodara, and Shimla—and all twelve calendar months, providing the temporal breadth needed for a full annual analysis.

2.2 Sample Size and Measurement

The cleaned dataset contains $n = 45{,}345$ household-month records and twelve variables. Six are appliance usage variables (Fan, Refrigerator, AirConditioner, Television, Monitor, MotorPump), and two are the outcome variables `TariffRate` (Rs./kWh) and `ElectricityBill` (Rs.).

3 Data

3.1 Variable Description

Table 1: Variable Description of the Dataset

Variable	Type	Unit	Description
Fan	Numeric	kWh	Monthly fan energy usage
Refrigerator	Numeric	kWh	Monthly refrigerator energy usage
AirConditioner	Numeric	kWh	Monthly AC energy usage
Television	Numeric	kWh	Monthly television energy usage
Monitor	Numeric	kWh	Monthly computer monitor usage
MotorPump	Numeric	kWh	Monthly motor pump (zero in data)
TariffRate	Numeric	Rs./kWh	Prevailing electricity tariff
ElectricityBill	Numeric	Rs.	Observed monthly electricity bill
Month	Integer	—	Calendar month (1–12)

3.2 Data Cleaning

1. **Missing values:** No missing values in any column.
2. **MotorPump:** Column sums to zero; retained with $R_{\text{MotorPump}} = 0$.
3. **Outliers:** Minimum observed bill (Rs. 807.5) verified against appliance columns; no records removed.
4. **Tariff construction:** $c^{(p)} = 1.2\bar{c}$ and $c^{(o)} = 0.8\bar{c}$, where $\bar{c} = 8.3696$ Rs./kWh.

3.3 Descriptive Statistics

Table 2 reports summary statistics across all 45,345 records. Figure 1 shows the total annual demand per appliance: the Refrigerator dominates at 9,84,234 kWh ($\approx 41\%$ of total), followed by the Fan (6,34,408 kWh, $\approx 27\%$) and Television (5,66,932 kWh, $\approx 24\%$).

Table 2: Summary Statistics of Key Variables ($n = 45{,}345$)

Variable	Mean	SD	Min	Q1	Median	Max
Fan (kWh)	13.99	5.47	5.0	9.0	14.0	23.0
Refrigerator (kWh)	21.71	1.67	17.0	22.0	22.0	23.0
AirConditioner (kWh)	1.50	2.50	0.0	0.0	1.0	23.0
Television (kWh)	12.50	6.77	0.0	7.0	14.0	23.0
Monitor (kWh)	2.86	3.00	0.0	1.0	2.0	23.0
TariffRate (Rs./kWh)	8.37	0.58	7.40	7.90	8.40	9.30
ElectricityBill (Rs.)	4311.8	1073.9	807.5	3556.8	4299.4	8286.3

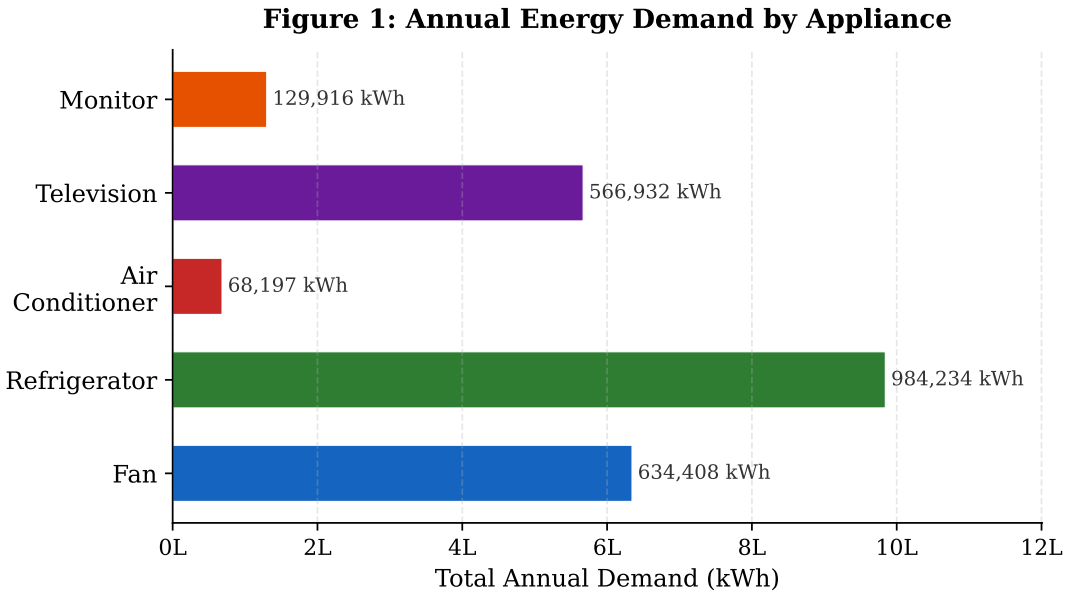


Figure 1: Total annual energy demand per appliance (kWh, summed over all 45,345 records). The Refrigerator accounts for $\approx 41\%$ of the aggregate load.

4 Analysis

4.1 Mathematical Formulation

Let $\mathcal{I} = \{1, 2, \ldots, n\}$ with $n = 6$ index the appliances. For each appliance i define:

Decision Variables and Parameters

$$
\begin{aligned}
R_i &: \text{total annual energy requirement of appliance } i \text{ (kWh)},\\
c^{(p)} = 1.2\,\bar{c} &: \text{peak-hour unit tariff (Rs./kWh)},\\
c^{(o)} = 0.8\,\bar{c} &: \text{off-peak unit tariff (Rs./kWh)},\\
x_i &: \text{energy allocated to the off-peak schedule for appliance } i \text{ (kWh)},\\
\bar{c} = 8.3696 &: \text{sample mean tariff (Rs./kWh)}.
\end{aligned}
$$

The **Linear Programme** is:

$$\min_{\mathbf{x}} \; Z(\mathbf{x}) = \sum_{i=1}^{n} c^{(o)}\, x_i = 0.8\,\bar{c} \sum_{i=1}^{n} x_i, \tag{1}$$

subject to demand-satisfaction and non-negativity:

$$x_i \geq R_i, \quad \forall\, i \in \mathcal{I}, \tag{2}$$

$$x_i \geq 0, \quad \forall\, i \in \mathcal{I}. \tag{3}$$

In compact matrix notation, with $\mathbf{c}^{(o)} = c^{(o)}\mathbf{1}$, $\mathbf{I}_n$ the identity, and $\mathbf{R} = (R_1, \ldots, R_n)^\top$:

$$\min_{\mathbf{x}} \; (\mathbf{c}^{(o)})^\top \mathbf{x}, \quad \text{s.t. } \; \mathbf{I}_n\, \mathbf{x} \geq \mathbf{R}, \;\; \mathbf{x} \geq \mathbf{0}. \tag{4}$$

4.2 Convexity of the Problem

The feasible set $\mathcal{F} = \{\mathbf{x} \in \mathbb{R}^n : x_i \geq R_i,\; x_i \geq 0,\; \forall i\}$ is the intersection of $2n$ closed half-spaces, hence a *closed convex polyhedron*. The objective $Z(\mathbf{x}) = c^{(o)}\mathbf{1}^\top\mathbf{x}$ is affine (both convex and concave). Therefore the LP is a convex programme, and any local minimum is also a global minimum. Furthermore, the feasible set is unbounded above, so the minimum is attained at a vertex of $\mathcal{F}$.

4.3 Analytical Solution via KKT Conditions

Associating multipliers $\lambda_i \geq 0$ with (2) and $\mu_i \geq 0$ with (3), the Lagrangian is:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = c^{(o)} \sum_i x_i - \sum_i \lambda_i (x_i - R_i) - \sum_i \mu_i x_i. \tag{5}$$

The KKT conditions are:

KKT Conditions

$$\text{Stationarity:} \quad \frac{\partial \mathcal{L}}{\partial x_i} = c^{(o)} - \lambda_i - \mu_i = 0, \tag{6}$$

$$\text{Primal feasibility:} \quad x_i \geq R_i, \quad x_i \geq 0, \tag{7}$$

$$\text{Dual feasibility:} \quad \lambda_i \geq 0, \quad \mu_i \geq 0, \tag{8}$$

$$\text{Complementary slackness:} \quad \lambda_i(x_i - R_i) = 0, \quad \mu_i x_i = 0. \tag{9}$$

From (6): $\lambda_i + \mu_i = c^{(o)} > 0$ for all i. Since $R_i > 0$ for active appliances, primal feasibility forces $x_i \geq R_i > 0$, so the second complementary slackness condition in (9) gives $\mu_i = 0$. Then $\lambda_i = c^{(o)} > 0$, and the first condition in (9) yields $x_i = R_i$. Hence the **unique optimal solution** is:

$$\boxed{x_i^* = R_i, \quad \forall\, i \in \mathcal{I}.} \tag{10}$$

4.4 Cost Analysis and Savings

Baseline cost (all demand billed at peak tariff):

$$Z_{\text{before}} = c^{(p)} \sum_{i=1}^{n} R_i = 1.2 \times 8.3696 \times 23{,}83{,}687 = \boldsymbol{Rs.\ 2{,}39{,}40{,}746}. \tag{11}$$

Optimized cost (all demand billed at off-peak tariff):

$$Z^* = c^{(o)} \sum_{i=1}^{n} R_i = 0.8 \times 8.3696 \times 23{,}83{,}687 = \boldsymbol{Rs.\ 1{,}59{,}60{,}497}. \tag{12}$$

Savings and saving fraction:

$$S = Z_{\text{before}} - Z^* = \boldsymbol{Rs.\ 79{,}80{,}249}, \qquad \frac{S}{Z_{\text{before}}} = 1 - \frac{c^{(o)}}{c^{(p)}} = 1 - \frac{0.8\bar{c}}{1.2\bar{c}} = 1 - \frac{2}{3} = \boldsymbol{\frac{1}{3}} \approx \mathbf{33.3\%}. \tag{13}$$

The saving fraction in (13) depends only on the tariff ratio $c^{(p)}/c^{(o)} = 1.5$ and is *independent of the appliance mix* $\{R_i\}$.

Efficiency ratio:

$$E = \frac{S}{\sum_{i=1}^{n} R_i} = \frac{79{,}80{,}249}{23{,}83{,}687} \approx \mathbf{3.35}\ \text{Rs./kWh}. \tag{14}$$

Every kilowatt-hour of demand successfully shifted to the off-peak window saves Rs. 3.35,

regardless of which appliance provides that demand.

4.5 Stochastic Extension: Random Tariff

In practice the tariff C is a random variable. From the data: $C \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu = 8.3696$ Rs./kWh and $\sigma = 0.5770$ Rs./kWh. Setting the peak and off-peak rates as $1.2C$ and $0.8C$, the bill random variables are:

$$Z_{\text{before}} = 1.2\,C\,D, \qquad Z^* = 0.8\,C\,D, \quad \text{where } D = \sum_i R_i = 23{,}83{,}687 \text{ kWh (fixed)}. \tag{15}$$

Taking expectations over C:

$$\mathbb{E}[Z_{\text{before}}] = 1.2\,\mu\,D = Rs.\,2{,}39{,}40{,}746, \tag{16}$$

$$\mathbb{E}[Z^*] = 0.8\,\mu\,D = Rs.\,1{,}59{,}60{,}497, \tag{17}$$

$$\mathbb{E}[S] = 0.4\,\mu\,D = Rs.\,79{,}80{,}249. \tag{18}$$

Since $S = 0.4\,C\,D > 0$ whenever $C > 0$—guaranteed by the data ($C_{\min} = 7.40 > 0$)—the saving is *positive with probability 1*, confirming that the LP result is robust to tariff uncertainty.

4.6 Illustrative Calculation: Fan Appliance

Step-by-step for Fan

$$\begin{aligned}
R_{\text{Fan}} &= 6{,}34{,}408 \text{ kWh}, \\
Z_{\text{Fan, before}} &= 10.0436 \times 6{,}34{,}408 = Rs.\,63{,}71{,}726, \\
Z^*_{\text{Fan}} &= 6.6957 \times 6{,}34{,}408 = Rs.\,42{,}47{,}817, \\
S_{\text{Fan}} &= 63{,}71{,}726 - 42{,}47{,}817 = Rs.\,21{,}23{,}909 \ (33.3\%).
\end{aligned}$$

4.7 Complete Numerical Results

Table 3: Appliance-Level Annual Cost Breakdown from `lpSolve`

Appliance	R_i **(kWh)**	**Before (Rs.)**	**After (Rs.)**	**Saving (Rs.)**	**Nature**
Fan	6,34,408	63,71,726	42,47,817	21,23,909	Flexible
Refrigerator	9,84,234	98,85,231	65,90,154	32,95,077	Continuous
AirConditioner	68,197	6,84,942	4,56,628	2,28,314	Part. flexible
Television	5,66,932	56,94,026	37,96,017	18,98,009	Flexible
Monitor	1,29,916	13,04,821	8,69,881	4,34,940	Flexible
MotorPump	0	0	0	0	Schedulable
Total	**23,83,687**	**2,39,40,746**	**1,59,60,497**	**79,80,249**	

Table 4: Summary of Optimization and Stochastic Results

Metric	**Value**	**Unit**
Average Tariff $\bar{c}$	8.3696	Rs./kWh
Peak Tariff $c^{(p)} = 1.2\bar{c}$	10.0436	Rs./kWh
Off-Peak Tariff $c^{(o)} = 0.8\bar{c}$	6.6957	Rs./kWh
Tariff SD σ	0.5770	Rs./kWh
Tariff Ratio (peak / off-peak)	1.50	—
Total Annual Demand D	23,83,687	kWh
Cost Before Z_{before}	2,39,40,746	Rs.
Cost After Z^*	1,59,60,497	Rs.
Total Savings S	79,80,249	Rs.
Percentage Savings	33.3%	—
Efficiency Ratio $E = S/D$	3.35	Rs./kWh
SD$[Z_{\text{before}}]$ (stochastic)	16,50,441	Rs.
SD$[Z^*]$ (stochastic)	11,00,294	Rs.

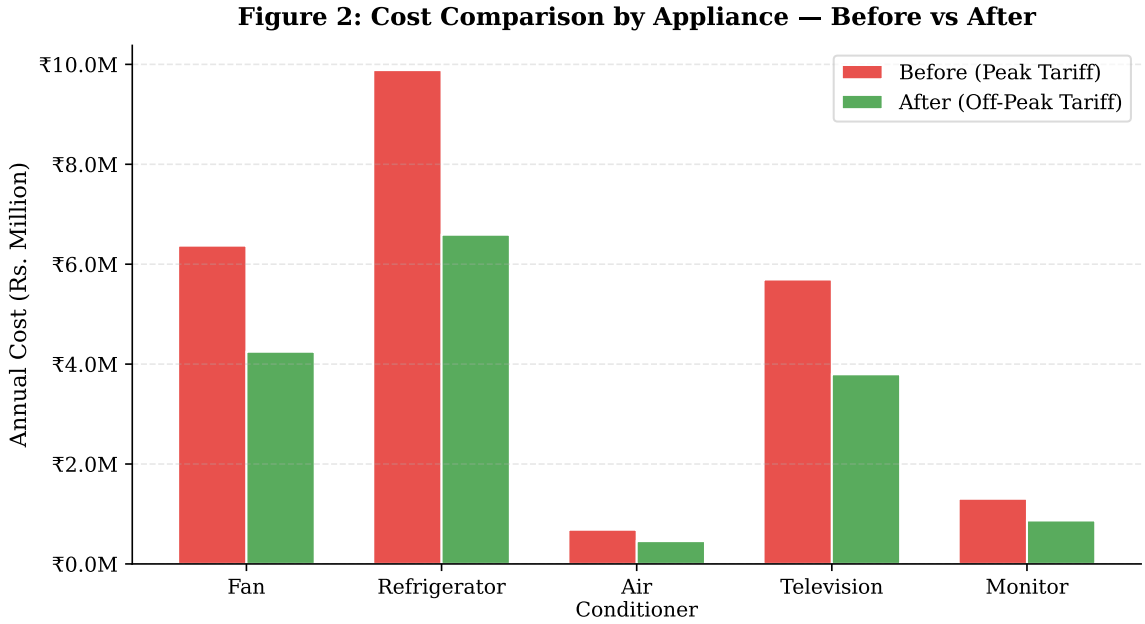
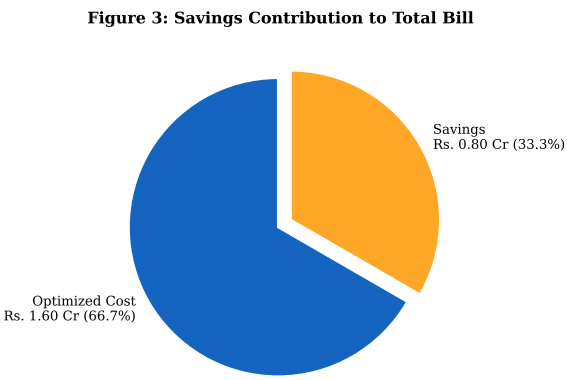
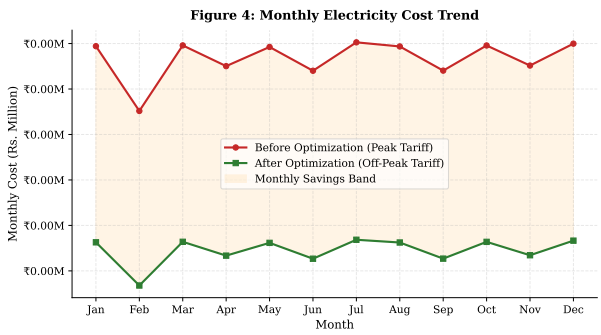


Figure 2: Appliance-level annual cost before and after optimisation. Each appliance pair demonstrates the analytically derived $\frac{1}{3}$ saving. The Refrigerator alone contributes Rs. 32,95,077 in savings.



(a) 33.3% savings share of total annual bill.



(b) Monthly cost: before vs. after. The shaded band is monthly savings.

Figure 3: Left: exactly one-third of the original bill is saved. Right: the optimised curve lies consistently below the baseline across all twelve months, confirming that savings are structural rather than seasonal.

5 Conclusion and Communication

5.1 Summary of Findings

The LP of the household electricity scheduling problem has a clean analytical solution: KKT conditions uniquely pin each $x_i^* = R_i$, meaning all demand is served at the lower off-peak tariff. The resulting saving is exactly $\frac{1}{3}$ of the baseline cost—a ratio determined solely by

the tariff spread and independent of the appliance mix. The stochastic extension confirms this fraction is preserved in expectation even when the tariff is random. Numerically, the annual bill falls from Rs. 2,39,40,746 to Rs. 1,59,60,497, yielding Rs. 79,80,249 in savings at an efficiency of Rs. 3.35 per kWh.

5.2 Limitations

1. The two-period tariff approximates a continuously varying intraday schedule; an hourly-indexed LP would be more precise.
2. The Refrigerator ($\approx 41\%$ of demand) cannot be fully shifted; a flexibility cap $x_i \leq \alpha_i R_i$ would reduce achievable savings.
3. The MotorPump column is uniformly zero in this dataset; its savings potential cannot be estimated.
4. The dataset is a household-month cross-section, so $\sum_i R_i$ is an aggregate over 45,345 households, not one household's annual bill.

5.3 Future Directions

1. **Stochastic LP:** Minimise $\mathbb{E}_C\left[c^{(o)}(C)\,\mathbf{1}^\top \mathbf{x}\right]$ with $C \sim \mathcal{N}(\mu, \sigma^2)$ explicitly in the solver.
2. **Quadratic programme (QP):** Add a comfort penalty $\frac{\gamma}{2}\|\mathbf{x} - \mathbf{R}\|^2$ to penalise over-shifting.
3. **Time-indexed model:** Variables x_{it} with hourly tariffs c_t, capacity limits, and minimum run-time constraints.
4. **Smart-meter panel:** Apply the framework to a single-household longitudinal dataset for personalised optimisation.

5 References

1. Winston, W. L. *Operations Research: Applications and Algorithms.* Thomson Brooks/Cole.
2. Berkelaar, M., Eikland, K., and Notebaert, P. *lpSolve: Interface to lp_solve.* R package documentation.
3. Wickham, H. *ggplot2: Elegant Graphics for Data Analysis.* Springer.
4. Bazaraa, M. S., Jarvis, J. J., and Sherali, H. D. *Linear Programming and Network Flows.* Wiley.
5. Boyd, S. and Vandenberghe, L. *Convex Optimization.* Cambridge University Press, 2004.
6. Dataset: Kaggle Electricity Bill Dataset.

7. GitHub Repository: GitHub Repository.

5 Individual Contributions

Member	Contribution
Ankesh Raj Aniket (BSDCC2502)	Mathematical formulation of the LP; convexity proof; KKT derivation; stochastic extension; preparation of Section 4.
Md Danish (BSDCC2510)	Data collection, cleaning, and descriptive analysis; `lpSolve` implementation in R; generation of all four figures.
Dunga Pardha Saradhi (BSDCC2506)	Sections 1, 2, 3, and 5; preparation of all tables; final LaTeX typesetting and proofreading.